

llmXive Automated Review of OpenThoughts-Agent: Data Recipes for Agentic Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a rigorous ablation study on data curation for agentic models, with most numerical claims aligning well with the provided tables. However, there are specific instances where the textual claims rely on unsupported or potentially hallucinated external facts, and one instance of ambiguous comparative logic.

First, the Introduction and Section 3.5 repeatedly reference “GPT-5.3-Codex” as a benchmark model and a teacher model. As of the current date, GPT-5 has not been publicly released, and no specific “5.3” version exists in the public record or the provided bibliography. The claim that it is the “best-performing model” is therefore unsupported by any cited evidence and appears to be a hallucination or a placeholder that was not updated. This undermines the credibility of the teacher model ablation results which hinge on this comparison.

Second, the claim in Section 3.5 that “GLM-4.7-AWQ is the best teacher” is ambiguous and potentially misleading when cross-referenced with Table 5. While GLM-4.7 achieves the highest *average* normalized score in that specific table (18.16), Kimi K2.5 achieves a higher raw average (18.59) and a significantly higher SWE-Bench score (33.33% vs 28.00%). The text justifies the claim by noting a ~5% decrease on Terminal-Bench 2.0 when using GPT-5.3, but it fails to acknowledge that Kimi K2.5 outperforms GLM-4.7 on the primary SWE-Bench metric. The “best teacher” label should be qualified (e.g., “best balanced teacher” or “best on average”) or the table should be re-evaluated to ensure the conclusion matches the data.

Finally, while the 30 pp and 10 pp impact claims in Section 3.1 are numerically supported by the range in Table 3, the 10 pp figure on Terminal-Bench is driven largely by the 0.37% outlier (AgentTuning-OS). It would be more accurate to specify that the range spans from the top strategy to the bottom outlier, rather than implying a typical variance of 10 pp across all strategies.

These issues are primarily matters of precision and verification rather than fundamental scientific flaws, but they require correction to ensure the claims are strictly supported by the evidence presented.

Action items.

- **[writing]** The paper presents a rigorous ablation study on data curation for agentic models, with most numerical claims aligning well with the provided tables. However, there are specific instances where the textual claims rely on unsupported or potentially hallucinated external facts, and one instance of ambiguous comparative logic. First, the Introduction and Section 3.5 repeatedly reference “GPT-5.3-Codex” as a benchmark model and a teacher model. As of the current date, GPT-5 has not been publicly rele

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively displays scaling trends, but the caption contains a broken reference to a table number and fails to define the ‘SERA’ methods shown in the legend, which are critical for interpreting the SotA claims.

Figure 2

The figure is a clear, readable pipeline diagram that matches its caption. The only issue is a placeholder (‘-’) in the caption where section numbers should be listed.

Figure 3

The figure is visually clear with a consistent legend, but it fails to render the error bars explicitly mentioned in the caption. Additionally, the textual claim that Method 1 plateaus is contradicted by the visual data in two of the three subplots.

Figure 4

The figure effectively visualizes the data pipeline flow, but the total count shown in the diagram (94,334) contradicts the caption’s claim of a 100k dataset. Additionally, the ‘StackExchange Tezos’ label suffers from poor formatting.

Figure 5

The figure effectively demonstrates scaling trends with clear error bars and baselines, but the legend labels do not explicitly match the specific model version (‘OpenThoughts-Agent-v2’) named in the caption, and the x-axis tick formatting is non-standard.

Figure 6

The figure effectively visualizes the reward collapse described in the caption, though the x-axis labels lack the year for clarity and there is a minor numerical discrepancy between the caption’s stated deployed reward (0.33) and the plotted marker position (~0.325).

Figure 7

The figure is generally clear but suffers from a legend entry being cut off by the plot border. Additionally, the position of the post-RL eval marker (orange diamond) appears inconsistent with the caption’s description of a successful, non-collapsing run reaching 0.73.

Figure 8

The figure effectively visualizes the temporal dynamics of the RL run, but the ‘Premature stop rate’ panel displays a constant value of 1.0 which contradicts the narrative of variable timeouts, and the x-axis labels are cluttered and overlapping.

Action items.

- **[writing]** Figure 1: The caption references ‘Table .’ with a missing table number, making the ‘All Average’ benchmark definition unresolvable.
- **[writing]** Figure 1: The legend includes ‘SERA Qwen3-32B (best harness)’ (dotted purple), but the caption states ‘Additional notes on SERA below’ without providing them, leaving the method undefined.
- **[science]** Figure 1: The ‘All Average’ plot shows a dotted purple line for ‘SERA Qwen3-32B (best harness)’ that is not present in the other two subplots, creating an inconsistent comparison set across the figure.
- **[writing]** Figure 2: The caption states ‘Each stage is ablated independently in Sections –’, but the section numbers are missing (likely a placeholder). Replace ‘–’ with the correct section numbers.
- **[science]** Figure 3: The caption states ‘Error bars are standard error across three stochastic re-runs,’ but the rendered plot contains no visible error bars on any data points.
- **[science]** Figure 3: The caption claims Method 1 ‘plateaus from 31.6K to 100K,’ yet the ‘SWE-bench Verified’ and ‘Terminal-Bench 2.0’ subplots show a visible increase in accuracy for Method 1 between these points, contradicting the description.
- **[science]** Figure 4: The Sankey diagram shows a total output of 94,334 (sum of Final Mix values), but the caption claims the final dataset is 100k agentic traces. The visual data does not support the specific number in the caption.
- **[writing]** Figure 4: The ‘StackExchange Tezos’ label is split across three lines, making it difficult to read and visually disjointed compared to other labels.
- **[science]** Figure 5: The legend labels the blue line as ‘OT-Agent Qwen3-8B’ and the pink line as ‘Nemotron-Terminal-Corpus Qwen3-8B’, but the caption states the experiment compares ‘OpenThoughts-Agent-v2’ against the ‘Nemotron-Terminal-Corpus baseline’. The legend fails to explicitly name the ‘OpenThoughts-Agent-v2’ method, creating ambiguity about whether the blue line represents the specific v2 recipe or a generic agent.
- **[writing]** Figure 5: The x-axis labels (‘316’, ‘1K’, ‘3.16K’, etc.) are not formatted as a standard logarithmic scale (e.g., 10^2 , 10^3) nor as a linear scale, which may confuse readers regarding the exact spacing of data points between 10K and 100K.
- **[science]** Figure 6: The caption states the deployed checkpoint reward is 0.33, but the ‘post-RL eval’ diamond marker is plotted at approximately 0.325, creating a minor inconsistency between the text and the visual data.

- **[writing]** Figure 6: The x-axis tick labels (e.g., ‘05-05 00’) lack the year, which is ambiguous for a scientific record; full ISO 8601 format (YYYY-MM-DD HH) is recommended.
- **[science]** Figure 7: The legend entry for the orange diamond (‘post-RL’) is partially cut off by the plot border, making the label illegible.
- **[science]** Figure 7: The caption claims the blue line rises from 0.54 to 0.73, but the visual data points start near 0.54 and end near 0.73; however, the orange diamond (post-RL eval) is plotted at ~ 0.22 , which contradicts the caption’s implication that the policy improves to 0.73 without collapse (the diamond should likely be higher if it represents the final eval reward).
- **[writing]** Figure 7: The legend text ‘post-RL eval’ is partially obscured by the plot frame on the right side.
- **[science]** Figure 8: The ‘Premature stop rate’ panel shows a constant value of 1.0 across all bins, which contradicts the caption’s claim that the agent ‘times out’ (implying a variable rate) and suggests a potential data logging error or mislabeled metric.
- **[writing]** Figure 8: The x-axis tick labels are rotated 45 degrees and overlap significantly, making the timestamps (e.g., ‘2026-05-05 00:00’) difficult to read.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally accessible to a competent reader from an adjacent field (e.g., an NLP researcher familiar with LLMs but not specifically with agentic RL pipelines). Most core concepts like SFT, RL, and standard benchmarks (SWE-Bench) are used correctly and are within the expected vocabulary of the discipline. However, there are several instances of in-group shorthand and undefined abbreviations that create minor friction for a reader not deeply embedded in this specific subfield’s current jargon.

Specifically, the use of “pp” for percentage points is a common statistical shorthand that should be expanded at first use to avoid ambiguity with other “parts per” metrics. The “Normalized” column in the ablation tables relies on a “z-score” definition mentioned in the text but not explicitly linked in the table headers, requiring the reader to hunt for the definition. Additionally, specific data source names like `\texttt{pymethods2test}` and model version strings like `GLM-4.7-AWQ` are used without brief operational glosses, assuming the reader knows the specific transformation or architecture implied by these names. Finally, the term “turns” in the context of agent rollouts could be clarified to distinguish between dialogue turns and tool-execution cycles.

These are minor barriers that can be resolved with simple parenthetical expansions or table caption clarifications, ensuring the paper is self-contained for the target audience.

Action items.

- **[writing]** Section 3.1 (Sourcing Tasks) and Table 1 use ‘pp’ (e.g., ‘~30 pp’) without defining it as ‘percentage points’. While common in economics/stats, an adjacent-field PhD in NLP/ML might momentarily confuse this with ‘parts per’ or ‘pages’. Define at first use: ‘percentage points (pp)’.
- **[writing]** Section 3.2 (Mixing Tasks) and Table 1 introduce ‘Raw’ and ‘Normalized’ columns without defining the normalization metric. The text mentions ‘average z-score’ in Section 3, but the table header does not explicitly link ‘Normalized’ to ‘z-score’, forcing the reader to cross-reference. Add ‘(z-score)’ to the ‘Normalized’ column header or define it in the table caption.
- **[writing]** Section 3.5 (Teacher Model) and Table 5 reference ‘GLM-4.7-AWQ’ and ‘GPT-5.3-Codex’. These specific versioned model names (especially the hypothetical/future ‘5.3’ and ‘4.7’) are used without a brief gloss of their architecture or origin (e.g., ‘GLM-4.7-AWQ, a quantized variant of the GLM-4.7 model’). While model names are proper nouns, the specific versioning implies a specific capability set not obvious to an outsider.
- **[writing]** Section 4.1 (Sourcing Tasks in RL) introduces the data source name ‘exttt{pymethods2test}’ without explaining the transformation. The text says it is ‘competitive programming recast as contracts’, but the name itself is opaque. Add a brief parenthetical on first use: ‘exttt{pymethods2test} (a dataset converting competitive programming problems into function-level contract verification tasks)’.
- **[writing]** Section 3.6 (Filtering Agent Rollouts) and Appendix A.3 use the term ‘turns’ (e.g., ‘fewer than 5 turns’) without explicitly defining it as ‘agent-environment interaction steps’ or ‘model response cycles’. In multi-turn dialogue, ‘turn’ is standard, but in agentic loops involving tool use, it can be ambiguous (does a tool call count as a turn?). Define ‘turn’ as ‘one complete cycle of model generation and tool execution’ at first use.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is generally sound, but specific textual claims do not strictly follow from the provided tabular data, creating logical gaps.

First, in Section 4.3 (“Filtering Agent Rollouts”), the text asserts that “Filtering traces with fewer than 5 turns yields the largest improvement.” However, Table 2 (e002) lists “Response Length (GPT-5, longest)” as the top strategy with a normalized score of +1.38, while the “min-turns” filter is not explicitly listed as a row in this table. The text’s claim of “largest improvement” is not entailed by the visible data in Table 2, which shows a different strategy as

the winner. The authors must either clarify that the min-turns filter corresponds to a specific row in the table or adjust the claim to reflect the actual top performer shown in the data.

Second, in Section 3.2 (“Mixing Tasks”), the text states that “Mixing top-ranked strategies (Top 4 or Top 8) outperforms the non-mixed Top 1 baseline.” While this is true for the “Normalized” and “Raw” average scores in Table 1 (e000), it is factually contradicted by the “SWE-Bench Verified (100)” column, where the Top 1 baseline (30.67%) outperforms the Top 4 mix (29.33%). Since SWE-Bench is a primary benchmark, the unqualified claim that Top 4 “outperforms” Top 1 is a non-sequitur relative to the specific SWE-Bench data presented. The claim should be qualified to “outperforms on average” or “outperforms on normalized score” to remain logically consistent with the table.

Finally, Section 5 (“Scaling Up SFT Data”) conflates two distinct scaling methods. It states that “Upsampling rollouts... plateaus at 31.6K” and then immediately discusses “Adding sources beyond Top-4... hurts performance.” While both statements are supported by their respective data points (Figure 2 for upsampling, Table 3 for source diversity), the narrative flow risks implying that the plateau in upsampling is the reason adding sources hurts, or that the “Top-16 hurts” result is a consequence of the upsampling plateau. The logic requires a clearer separation: Method 1 (upsampling) plateaus, while Method 4 (adding sources) has diminishing returns/harms performance independently. The current phrasing creates a slight causal ambiguity.

Action items.

- **[science]** Section 4.3 claims filtering for <5 turns yields the “largest improvement,” but Table 2 (e002) shows “Response Length (GPT-5, longest)” as the top strategy (+1.38 normalized) with no row for the min-turns filter. The text’s superlative claim is not entailed by the visible table data.
- **[science]** Section 3.2 states mixing Top 4/8 “outperforms” Top 1, but Table 1 (e000) shows Top 1 (30.67%) beats Top 4 (29.33%) on SWE-Bench Verified. The unqualified claim contradicts the primary metric column; qualify as “outperforms on average” or “normalized score.”
- **[writing]** Section 5 conflates “Upsampling” (Method 1) plateauing with “Adding sources” (Method 4) hurting performance. The text implies a single narrative for both, but the data (Fig 2 vs Table 3) supports them as distinct phenomena. Clarify the distinction to avoid causal ambiguity.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several broad claims about the generalizability and superiority of its “data recipe” that exceed the scope of the evidence, which is restricted to a single base model family

(Qwen3) and specific benchmark subsets.

First, the Conclusion and Introduction assert that OpenThoughts-Agent-v2 is the “strongest open-data $\leq 32\text{B}$ model on average across seven benchmarks.” While the average score in Table 2 (e002) is indeed the highest, the model significantly underperforms the Nemotron-Terminal-32B baseline on MedAgentBench (47.8% vs 62.6%). The rhetoric of being the “strongest... across seven benchmarks” risks implying a more uniform dominance than the data supports. The claim should be qualified to reflect that the model leads on average, particularly on software engineering tasks, but does not universally outperform all baselines on every single benchmark listed.

Second, the paper frames its findings as a general “data recipe” for agentic models. However, the entire experimental suite (SFT and RL) is conducted exclusively on the Qwen3-8B and Qwen3-32B backbones. The Conclusion admits “no ablation of base model choice” as a limitation, but this is a fundamental boundary on the validity of the “recipe” claim. The paper does not demonstrate that the specific mixing strategies, filtering heuristics, or RL sources generalize to other model families (e.g., Llama or Mistral). The rhetoric should be tightened to specify that these are “Qwen3-optimized data recipes” or explicitly state that generalization to other architectures remains an open question, rather than presenting the findings as a universal solution for agentic data curation.

Finally, the Introduction claims the dataset “outperforms prior work” on a list of OOD benchmarks. While true for most, the omission of the MedAgentBench failure in the narrative (where it loses by a significant margin) creates a cherry-picked impression of robustness. A more honest framing would acknowledge that while the method improves the average, it does not guarantee superiority on every domain, particularly those outside the primary software engineering focus.

Action items.

- **[writing]** Conclusion claims ‘strongest... across seven benchmarks’ despite losing on MedAgentBench (47.8% vs 62.6%). Narrow to ‘strongest on average’ or specify the subset of benchmarks where it leads to avoid implying universal dominance.
- **[writing]** Introduction claims outperformance on listed OOD benchmarks but omits the MedAgentBench failure. Add a limitation acknowledging the model does not dominate every single OOD benchmark to prevent a cherry-picked impression of robustness.
- **[writing]** Conclusion admits ‘no ablation of base model choice’ but presents findings as a general ‘data recipe.’ Explicitly state that the recipe is validated only on Qwen3 and generalization to other families (e.g., Llama) is unproven.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a data curation pipeline for training agentic language models on software engineering and terminal tasks. The work involves supervised fine-tuning (SFT) and reinforcement learning (RL) using synthetic and public benchmark data (e.g., SWE-Bench, Terminal-Bench).

From a safety and ethics perspective, the research is low-risk by construction. The primary artifacts are data recipes and model weights trained on public or synthetic datasets. The paper explicitly acknowledges the dual-use nature of agentic models in the “Broader Impacts” section (Section 6), noting that these models are “capable of unauthorized actions” and recommending “sandboxing and human oversight.” This disclosure is appropriate for the field and the specific capabilities described.

There are no indications of:

- **Human subjects data:** The data sources are synthetic (generated by other models) or derived from public benchmarks (SWE-Bench, Terminal-Bench) which do not involve private human data or require IRB approval in this context.
- **PII exposure:** The paper does not release raw scraped data containing personal information; it releases processed trajectories and model weights.
- **Dual-use operationalization:** While the models can perform coding tasks, the paper does not provide operational details for generating exploits, malware, or specific cyber-attacks. The focus is on general software engineering capabilities (fixing bugs, implementing features).
- **License violations:** The paper cites the sources of its data (e.g., SWE-Smith, StackExchange) and releases its own artifacts under open licenses, with no evidence of ToS violations in the text provided.

The paper does not require additional safety disclosures or mitigations beyond what is already present. The risk profile is consistent with standard ML research on agentic systems.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents an extensive ablation study on data curation for agentic models, but several central claims rely on experimental designs that do not fully rule out confounding variables or sampling noise.

First, the stability of the reported gains is unclear. Tables 1, 2, and 3 report standard errors (e.g., ± 1.63) but do not state the number of random seeds (n) used to generate these statistics. In agentic benchmarks, run-to-run variance can be high; a 1.3 percentage point difference (e.g., Top 4 vs. Top 2 mixing in Table 1) is well within the margin of error for a single seed. Without disclosing the seed count (e.g., $n=3$ or $n=5$), the reader cannot distinguish a robust effect from

lucky initialization.

Second, the “compute-controlled” claims in the filtering and scaling sections are not fully supported by the presented tables. Section 3.6 asserts that filtering for longer traces improves performance “at a matched token budget,” yet the provided tables (Table 2, Appendix A.3) do not explicitly display the control run where token budget is held fixed while turn count varies. Similarly, the scaling analysis in Section 4 attributes gains to data volume and augmentation, but Table S4 reveals that the hyperparameters (epochs and gradient clipping) change simultaneously with data scale (e.g., 7 epochs for 10K data vs. 5 epochs for 100K data). This confounds the effect of data diversity with the effect of training duration and optimization dynamics.

Finally, the RL conclusion that “undertrained” models benefit more is based on a comparison between a 10K SFT model and a 100K SFT model. However, as noted above, these models also differ in training epochs (7 vs. 5). The observed benefit could be an artifact of the 10K model receiving more training steps rather than a specific property of being “undertrained” in terms of data volume.

To close these gaps, the authors should: (1) explicitly state the number of seeds for all ablation runs; (2) provide the specific compute-controlled ablation tables for the filtering and scaling experiments where hyperparameters are held constant; and (3) run a control experiment for the RL comparison where the SFT models at different scales are trained for the same number of epochs.

Action items.

- **[writing]** The paper presents an extensive ablation study on data curation for agentic models, but several central claims rely on experimental designs that do not fully rule out confounding variables or sampling noise. First, the stability of the reported gains is unclear. Tables 1, 2, and 3 report standard errors (e.g., ± 1.63) but do not state the number of random seeds (n) used to generate these statistics. In agentic benchmarks, run-to-run variance can be high; a 1.3 percentage point difference (e.g., T

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical reporting in this paper is generally transparent regarding the magnitude of effects but lacks rigor in inferential claims and uncertainty quantification.

Uncertainty Reporting: Tables 1, 2, and 3 report point estimates with subscripts (e.g., 29.33 ± 1.63). While the caption for Figure 2 mentions “standard error across three stochastic re-runs,” the ablation tables (which cover 95 strategies) do not explicitly define these subscripts as Standard Error (SE) or Standard Deviation (SD), nor do they consistently state the sample size (N) for every experiment. Given the high variance in LLM agent performance, distinguishing between SE (precision of the mean) and SD (variability of the population) is critical for

interpreting whether a 1-2 point difference is meaningful. The authors should add a footnote to all tables defining the subscript and explicitly stating N.

Significance vs. Observation: The text frequently uses phrases like “improves performance by ~3 pp” or “outperforms baselines” in the context of ablation studies. However, no hypothesis tests (t-tests, Wilcoxon) or confidence intervals are reported for these specific pairwise comparisons. With N=3 (as hinted in the scaling section), statistical power is low, and observed differences could easily be noise. The authors should either run the appropriate paired tests for the key ablation comparisons (e.g., Top-4 vs. Top-1) and report p-values, or soften the language to “observed improvement” to avoid implying statistical significance where none was tested.

Multiple Comparisons: The study ablates 95 task generation strategies and several mixing/filtering combinations. The paper highlights the “top” performers (e.g., SWE-Smith, Top-4 mixing) as the definitive best choices. Without a correction for multiple comparisons (e.g., Holm-Bonferroni or Benjamini-Hochberg), the probability of at least one false positive among 95 tests is extremely high. The authors should apply a correction to the ablation results or explicitly acknowledge that the “best” strategies are selected from a large pool and may not be robust to the multiple testing problem.

Precision: Results are reported to two decimal places (e.g., 29.33%). Given the small N (likely 3) and the inherent stochasticity of agent rollouts, this level of precision is misleading (false precision). Reporting to one decimal place or using ranges would be more appropriate.

Action items.

- **[writing]** Table 1 (mixing_strategies) and Table 2 (task_gen_full) report means with subscripts (e.g., 29.33 ± 1.63) but do not define the subscript as Standard Error (SE) or Standard Deviation (SD). Section 4.2 mentions ‘three stochastic re-runs’ for scaling plots, but the ablation tables (95 strategies) lack explicit confirmation of N. Report ‘N’ and the specific metric (SE vs SD) in table captions to allow readers to judge precision and perform meta-analysis.
- **[writing]** The paper claims ‘significant’ improvements (e.g., +3 pp avg in filtering, +5.4 pp on SWE-Bench) without reporting p-values or confidence intervals for these specific pairwise comparisons. With N=3 runs, a paired t-test or Wilcoxon signed-rank test is feasible. Either report the test statistic and p-value for the key ablation comparisons or rephrase claims to ‘observed improvement’ to avoid implying statistical significance without evidence.
- **[science]** Section 4.2 and Table 3 present results from 95 task generation strategies and multiple mixing strategies. The ‘best’ performers are highlighted without correction for multiple comparisons (e.g., Bonferroni or FDR). With ~100+ tests, false positives are expected. Apply a multiple-comparison correction to the ablation results or explicitly state that the reported ‘best’ strategies are uncorrected and subject to selection bias.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured, with clear section headings and a logical progression from data curation to scaling and reinforcement learning. However, several specific instances of ambiguous phrasing and potential logical contradictions in the text impede immediate comprehension.

First, in Section 3.5 (“Teacher Model”), the sentence regarding the performance decrease when using GPT-5.3-Codex is ambiguous. The phrase “representing a ~5% decrease” could be interpreted as a relative percentage drop or an absolute percentage point drop. Given the context of benchmark scores, the latter is likely intended, but the phrasing is imprecise and forces the reader to guess. This should be clarified to “a 5 percentage point decrease” or similar.

Second, there is a likely contradiction in Section 3.6 (“Filtering Agent Rollouts”). The text states, “Filtering traces with fewer than 5 turns yields the largest improvement.” This contradicts the general intuition of the paper (and the results in Section 3.4 regarding token counts) that longer, more complex traces are beneficial. It is highly probable that the author meant “more than 5 turns” or “at least 5 turns.” This typo creates a significant friction point where the reader must pause to reconcile the text with the apparent logic of the experiment.

Third, the Introduction’s bulleted list of findings includes a claim about teacher models that is not immediately grounded in the surrounding text. While the claim is valid based on the body, the introduction would benefit from a brief mention of the specific models compared (GLM vs. GPT) to make the finding concrete for a skimming reader.

Finally, the transition into the “OpenThoughts-Agent-v2” results paragraph in Section 4 is abrupt. The paragraph jumps straight into metrics without a sentence connecting the previous discussion on scaling methods to these final numbers. A simple signposting sentence would smooth this handoff.

Addressing these specific points will ensure the reader can move through the paper without having to re-read or guess the author’s intent.

Action items.

- **[writing]** Section 3.5 (Teacher Model) states: ‘Despite GPT-5.3-Codex being the best-performing model, GLM-4.7-AWQ is the best teacher, representing a ~5% decrease in performance when using GPT-5.3-Codex.’ The phrase ‘representing a ~5% decrease’ is ambiguous: does it mean a 5 percentage point drop or a 5% relative drop? Rewrite to specify the metric (e.g., ‘a 5 percentage point drop on Terminal-Bench 2.0’).
- **[writing]** Section 3.6 (Filtering Agent Rollouts) claims: ‘Filtering traces with fewer than 5 turns yields the largest improvement.’ This contradicts the logic of Section 3.4 (Filtering Tasks), which found that filtering for *more* tokens (implying longer traces) improved performance. The phrasing ‘fewer than 5 turns’ likely contains a typo (should be ‘more than’ or ‘at least’). Verify and correct the direction of the filter to match the reported results.
- **[writing]** The Introduction lists four key findings in a bulleted list, but the second bullet (‘The strongest model by benchmark performance does not necessarily make the best teacher’) is not explicitly supported by a specific result summary in the abstract or intro text. Ensure

the intro explicitly names the teacher model comparison (GLM vs. GPT) to ground the claim before the reader reaches Section 3.5.

- **[writing]** Section 4 (Scaling Up SFT Data) contains a paragraph labeled ‘OpenThoughts-Agent-v2’ that abruptly lists final metrics without a transition sentence explaining how these results relate to the preceding discussion on scaling methods. Add a lead-in sentence (e.g., ‘Applying these scaling strategies to the full 100K dataset yields the following final performance:’) to improve flow.